

Outlets Tab - Stage/Storage/Discharge Dialog Box

Use this tab to enter the stage-storage-discharge data to specify the discharge values that correspond to the stage values.

Culverts / Orifice

Rise (in)

Specifies the diameter of the outlet pipe or the height of the box section.

Span (in)

Specifies the diameter of the outlet pipe or the width of the box section. **This is a required entry.**

NOTE:

If the Rise and Span values are equal, Hydraflow Hydrographs Extension assumes a circular section. If they are unequal by any amount, then Hydraflow Hydrographs Extension assumes a box or rectangular section.

No. Holes

Specifies the total number of orifices in the perforated riser. For more information, see [Perforated Risers](#).

No. Barrels

Specifies the number of barrels or pipes for the structure.

Invert Elev. (ft)

Specifies the elevation of the invert at the upstream end.

Height (ft)

Specifies the distance from the invert of the lowest hole to the top of the highest hole or the center of the lowest orifice of perforated riser to the center of the highest orifice plus the S-value. For more information, see [Perforated Risers](#).

Length (ft)

Specifies the length of structure. If you are modeling an orifice, enter 0.

Slope (%)

Specifies the slope of the culvert in percent. If you are modeling an orifice, enter 0.

N-Value

Specifies Manning's n-value.

Orifice Coeff.

Specifies the orifice coefficient. The default value is 0.60.

Multi-Stage

Specifies the model as a multi-stage structure. Use only for orifice B, C, and Perf. Riser. Multi-stage structure implies that all of the flows from this structure passes through culvert A.

Active

Specifies the structure as an active component in the final stage-discharge calculation. When testing various scenarios and structures, you can use this control to activate some structures and deactivate others.

Tailwater

Tailwater Elevation (ft)

Specifies the elevation of the HGL (hydraulic grade line) at the downstream end.

Weirs

Weir Type

Specifies the type of the weir: Rectangular, Cipoletti, Broad crested, Riser, or V-Notch (from 5 to 120 degrees).

NOTE:

Only weir A can be a structure of type Riser.

Crest Elev (ft)

Specifies the elevation of the weir crest.

NOTE:

Crest elevation for V-Notch weirs is the bottom of the weir, not the top.

Crest Length (ft)

Specifies the length of the weir crest. For a standpipe weir, the length is equal to the circumference of the weir. For V-Notch weirs, the length is 0.

Weir Coeff.

Specifies a weir coefficient. Hydraflow Hydrographs Extension automatically selects the default coefficient, which is based on the selected weir type. It is recommended that you use the default values. See [weir calculation methods](#)

Multi-Stage

Specifies whether or not you are modeling a multi-stage structure. For more information, see [About Multi-Stage Structures](#).

Active

Specifies the structure as an active component in the final stage-discharge calculation. When testing various scenarios and structures, you can use this control to activate some structures and deactivate others.

Exfiltration

Contains data to be added to the outflow exfiltration component of your structure.

Rate (in/hr)

Specifies the exfiltration rate.

Apply to

Specifies a type of area to which apply exfiltration data: Contour or Wetted Surface Area.

Extract from Outflow Hyd (y/n)

Specifies the removal of the exfiltration flows from any hydrograph routed through the current pond. If you select No, the outflow hydrograph contains these flows, and it may be necessary to divert the exfiltration component before further downstream processing.

Compute button

Click to update the table.

Stage-storage-discharge table

Contains the computed discharges at each stage and elevation for each outlet structure, as well as the total discharge. Specify the known discharges at each stage in the User-defined column. The values are then added to the total.